

VWAP

VWAP is the volume weighted average price during the trading period.

$$VWAP = \frac{\sum_{i=1}^t p_i \cdot v_i}{\sum_{i=1}^t v_i} \quad (5.8)$$

First Price

The mid-point of the bid-ask spread at the beginning of the trading interval. This is denoted as P_0 .

Average Daily Volume

The average daily traded volume (ADV) in the stock over the previous T trading days. The value of T does vary by practitioner. For example, the more common historical periods are 10, 22, 30, and 66 days of data. Earlier we found that $T = 30$ days of data are a sufficient number of data points to measure the mean.

$$ADV = \frac{1}{T} \sum_{i=1}^T V_i(day) \quad (5.9)$$

where $V_i(day)$ is the total volume that traded on the i^{th} historical day (e.g., i days ago).

Annualized Volatility

Annualized volatility is the standard deviation of the close-to-close logarithmic price change scaled for a full year using a factor of 250 days. Many practitioners use a 252 day scaling factor. However, for our purposes estimating market impact, the difference is negligible. Annualized volatility is included in the market impact model as a proxy for price volatility. For consistency, we use $T = 30$ days of data to compute our volatility estimate.

$$\sigma = \sqrt{\frac{250}{T-1} \sum_{i=2}^T (r_i - r_{avg})^2} \quad (5.10)$$

where, r_i is the log return on the i^{th} historical day and r_{avg} is the average log return over the period. It is important to note that our annualized volatility is expressed as a decimal (e.g., $0.20 = 20\%$).